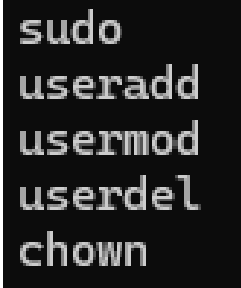


Managing Users in Linux Lab

Summary: Understanding how to add and manage users on a shared system is incredibly useful when it comes to Linux environments. I am tasked with adding a new user and giving them the appropriate access to the organization. The goal of this lab is to get more comfortable handling user accounts to further protect an organization's system and ensure only authorized users have access.

Commands Used:



```
sudo
useradd
usermod
userdel
chown
```

Overview: The first thing I needed to do was add the new user using `sudo useradd` followed by the username. Using the `sudo` command is required because creating or modifying accounts is an administrative task that needs elevated privileges. After creating the account, I assigned the user to their primary group with the `usermod -g` command, so they were correctly associated with the research group.

Next, I needed to update the ownership of a specific project file so the new employee could take responsibility for it. I used the `chown` command to transfer ownership of the file to the new user. This ensured they had the correct level of control over the resource without granting unnecessary access to others. Later in the scenario, the employee's responsibilities expanded, so I added them to a secondary group using `usermod -a -G`. The `-a` option appends the user to additional groups without removing them from their existing ones, and the `-G` option specifies which supplementary group to add them to. This allows them to remain part of their original group while gaining access to resources used by the sales team.

Eventually, the employee left the organization, so I removed their account with `sudo userdel` and cleaned up the empty group that was automatically created for them. This prevents unused accounts and groups from lingering on the system, which is an important part of maintaining security.

Reflection: This lab emphasized how user management directly supports system security. Adding, modifying, and removing accounts reinforced how important it is to keep access aligned with a user's current role. This scenario demonstrates how proper account maintenance helps prevent unauthorized access and keeps a system organized as people join, move between teams, or leave an organization.